

Sentiment Analysis Using Natural Language Processing (NLP)

Codveda Technology – Data Analytics Internship

Submitted By

Name: Pawan Kushwah

College: G H Rasoni International Skill Tech University, Pune

Branch: B.Tech Artificial Intelligence & Data Science

Objective

The objective of this project is to analyze text data using Natural Language Processing techniques and classify sentiments expressed in social media posts.

Dataset

Dataset Name: Sentiment Dataset

The dataset contains social media posts along with their corresponding sentiment labels such as Positive, Negative, and Neutral.

Tools Used

- Python
 - Google Colab
 - Pandas
 - NumPy
 - NLTK
 - Matplotlib
 - Seaborn
 - WordCloud
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Methodology

Step 1 – Data Loading

The dataset was loaded using Pandas.

Step 2 – Data Cleaning

- Missing values were checked.
- Required columns were selected.
- Text was converted to lowercase.
- URLs, punctuation, and extra spaces were removed.

Step 3 – Text Preprocessing

- Tokenization
- Stopword Removal

Step 4 – Visualization

- Sentiment Distribution
- Word Cloud

Step 5 – Data Preparation

The processed text was prepared for future NLP and machine learning tasks.

Results

- Text data was successfully cleaned and preprocessed.
 - Tokenization and stopwords removal improved text quality.
 - Sentiment distribution was visualized using a count plot.
 - Frequently occurring words were visualized using a Word Cloud.
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Conclusion

Natural Language Processing techniques were successfully applied to preprocess and analyze text data. The project demonstrates how sentiment analysis can be used to understand public opinion and prepare textual data for machine learning applications.